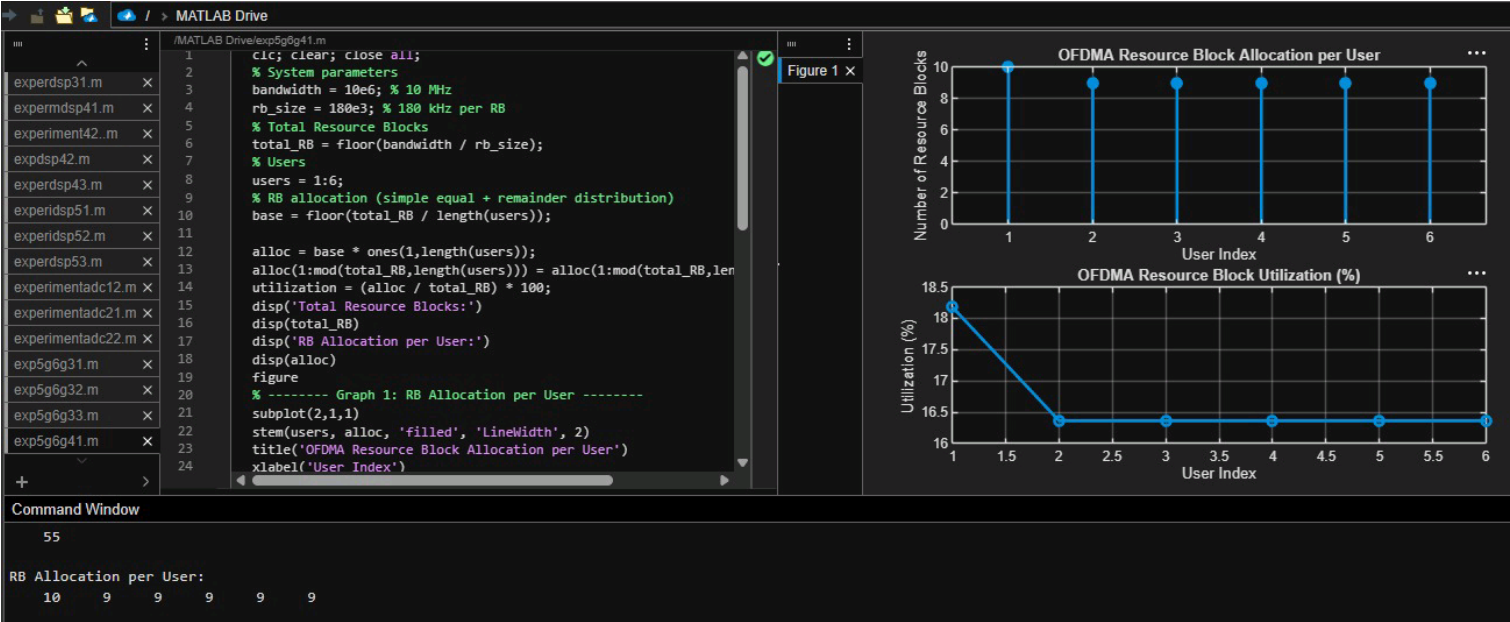
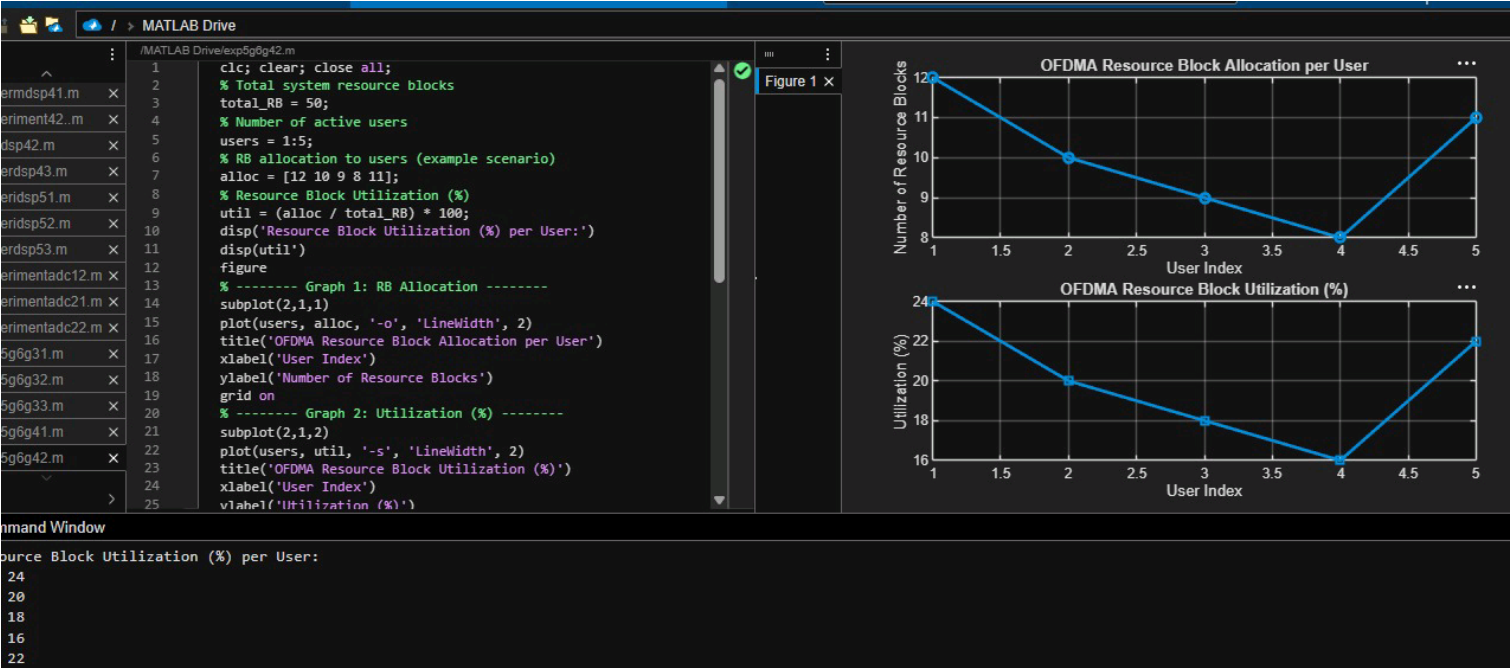






MATLAB Drive

experiment42.m

expdsp42.m

expersp43.m

expersp51.m

expersp52.m

expersp53.m

experimentadc12.m

experimentadc21.m

experimentadc22.m

exp5g6g31.m

exp5g6g32.m

exp5g6g33.m

exp5g6g41.m

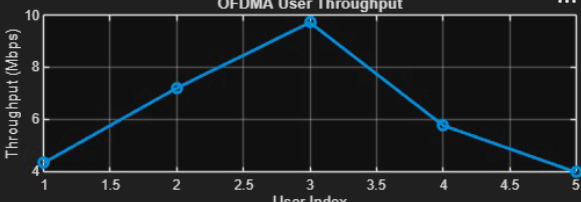
exp5g6g42.m

exp5g6g43.m

```
3 rb_bw = 180e3; % Bandwidth per RB (180 kHz)
4 users = 1:5;
5 % Allocated Resource Blocks per user
6 alloc_RB = [12 10 9 8 11];
7 % Modulation efficiency (bits/symbol)
8 mod_eff = [2 4 6 4 2]; % QPSK, 16QAM, 64QAM etc.
9 % Throughput calculation (Mbps)
10 throughput = (alloc_RB .* rb_bw .* mod_eff) / 1e6;
11 disp('User Throughput (Mbps):')
12 disp(throughput)
13 figure
14 % ----- Graph 1: Throughput per User -----
15 subplot(2,1,1)
16
17 plot(users, throughput, '-o', 'LineWidth', 2)
18 title('OFDMA User Throughput')
19 xlabel('User Index')
20 ylabel('Throughput (Mbps)')
21 grid on
22 % ----- Graph 2: RB vs Throughput -----
23 subplot(2,1,2)
24 plot(alloc_RB, throughput, '-s', 'LineWidth', 2)
25 title('Resource Blocks vs Throughput Relationship')
26 xlabel('Allocated Resource Blocks')
27 ylabel('Throughput (Mbps)')
```

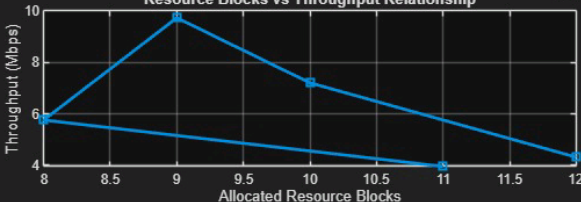
Figure 1

OFDMA User Throughput



User Index	Throughput (Mbps)
1	4.3200
2	7.2000
3	9.7200
4	5.7600
5	3.9600

Resource Blocks vs Throughput Relationship



Allocated Resource Blocks	Throughput (Mbps)
12	4.3200
10	7.2000
9	9.7200
11	5.7600
8	3.9600

Command Window

User Throughput (Mbps):
4.3200
7.2000
9.7200
5.7600
3.9600